Juan Esteban Mayoral Zepeda

Mexico City, Mexico
juanestebanmayoral@gmail.com | +52-5549181189 | LinkedIn | GitHub

Professional Summary

AI and Data Engineer specialized in building scalable backend systems, data pipelines, and AI-powered applications. Experienced implementing distributed data processing, feature-rich APIs, vector-based retrieval systems, and intelligent services using modern cloud-native architectures.
Proficient in designing and deploying data workflows, APIs, and AI components using **Python, FastAPI, SQL, Spark, Docker, Kubernetes, AWS, and event-driven systems**. Skilled in developing high-performance services for data ingestion, transformation, analytics, and model-driven features.
Hands-on expertise with modern AI technologies such as **LangChain, LangGraph, vector databases, LLM orchestration patterns, and retrieval-augmented generation (RAG)**. Experienced integrating LLMs into production systems with secure inference layers, optimized routing, and robust backend logic.
Combines strong backend engineering with deep AI development experience, enabling collaboration across data, infrastructure, and software teams. Passionate about building reliable, intelligent, and maintainable systems powered by next-generation AI technologies.

Technical Skills

Programming (Python, SQL, TypeScript)
AI Engineering (LangChain, LangGraph, RAG)
AWS Cloud (Lambda, S3, DynamoDB, Bedrock)
Data Engineering (Spark, ETL Workflows)
DevOps (Docker, Kubernetes, Terraform)
Backend (FastAPI, Django, Microservices)
AI Infra (Vector DBs, Kong AI Gateway)
Monitoring & Automation (CloudWatch, Grafana)
Databases (PostgreSQL, Redis)
Developer Tools (Linux, Git, CI/CD)

Professional Experience

Vyvo AI — *AI Lead Engineer | 2025 – Present*

- Built and deployed AI-powered services using **LangChain, LangGraph, vector databases, and RAG architectures**, enabling real-time personalization and intelligent decision flows.
- Developed and operated LLM inference layers secured with **Kong AI Gateway**, implementing rate limiting, JWT validation, logging hooks, and cost-optimized inference paths.
- Designed backend logic and data flows for AI-driven features across mobile and backend teams, improving reliability and integration speed of AI components.
- Collaborated with data and engineering teams to establish architectural patterns for AI services, vector-based retrieval, and scalable inference workloads.

Nyx Technology — *Cloud Engineer | Oct 2024 – Jul 2025*

- Contributed to the end-to-end development of **Flowlink**, a Channel Manager platform deployed for Marriott/City Express, automating and synchronizing rates, availability, and reservations across multiple OTAs, reducing operational overhead by over **120%**.
- Engineered a production-grade backend using **FastAPI, AWS Lambda, PostgreSQL (RDS), and S3**, achieving a highly modular and scalable system with **99.9% availability**.
- Spearheaded the deployment and customization of a Moodle-based LMS for StackUP University, integrating third-party APIs and automating onboarding, grading, and course provisioning for **1,000+ users**.
- Designed and delivered multiple **end-to-end AI/ML PoCs for Alsea and OXXO**, implementing complete AWS pipelines using **Lex, Bedrock, Personalize, Transcribe, and Textract**. Led infrastructure provisioning, backend development, and model integration, enabling production-ready prototypes—one of which was deployed to stores for real operational use.

Universidad Panamericana — *Solutions Architect | Jan 2023 – Sept 2024*

- Led architectural design for multiple backend and mobile systems, defining domain boundaries, integrations, and scalable API standards across iOS and backend teams.
- Designed and implemented RESTful APIs, event workflows, and data pipelines using **FastAPI, PostgreSQL, Redis, and AWS serverless components**.
- Re-architected an AI-powered chatbot system, migrating to modular inference pipelines and optimizing compute usage, reducing operational AI costs by **40%**.

- Conducted system-wide audits identifying performance bottlenecks, security gaps, and scalability issues, delivering architectural improvements across the platform.

Freelance Developer *Jun 2021 – Nov 2022*

- Designed and deployed a conversational bot using the **Twilio WhatsApp API**, automating reservation management for Airbnb listings and decreasing average response time from **5 minutes to 30 seconds**.

Projects & Achievements

AI/ML PoCs for Alsea & OXXO (AWS Lex, Bedrock, Transcribe, Textract, Personalize)

- Designed and delivered multiple **end-to-end AI/ML proof-of-concepts** for enterprise clients Alsea and OXXO, integrating AWS-native AI services to enable automation, personalization, and intelligent document/voice processing.
- Built complete cloud pipelines using **Lex, Bedrock, Personalize, Transcribe, Textract**, handling infrastructure provisioning, backend logic, and orchestration with AWS Lambda and API Gateway.
- Delivered production-ready prototypes; one solution was deployed into **OXXO retail stores** for real operational testing and feedback.

UP Intelligent Chatbot – AI Architecture (FastAPI, AWS, Redis, LangChain)

- Designed the full architecture for the Universidad Panamericana's intelligent chatbot, integrating vector retrieval, embedding models, and modular inference flows.
- Developed knowledge-base ingestion, preprocessing utilities, and search pipelines using **LangChain**, improving reasoning accuracy and response quality.
- Deployed scalable inference and API layers using AWS serverless components, enabling fast updates and low-latency responses.

Education

Universidad Panamericana *Mexico City, Mexico*
Data Intelligence & Cybersecurity

Certifications

- **AWS Certified AI Practitioner** Jul 2025
- **AWS Certified Cloud Practitioner** Aug 2024